

Amazon Data Engineer Interview Questions

(Most Frequently Asked)

This document contains real and frequently asked questions in Amazon Data Engineer interviews, categorized by topics. Each section includes descriptive details to help you prepare effectively.

1. SQL & Data Modeling

- Write a SQL query to find the second highest salary from the Employee table.
- How do you design a schema for storing IoT sensor data at scale?
- Explain the difference between OLTP and OLAP systems with examples.
- How do you handle slowly changing dimensions (SCD Type 2) in a data warehouse?
- Given two large tables, how would you optimize a join in Amazon Redshift?

2. Data Warehousing & ETL

- What are the differences between Amazon Redshift and Snowflake?
- How would you optimize ETL jobs in AWS Glue?
- How do you implement partitioning and bucketing in Hive or Spark?
- What are the best practices for designing a star schema in Redshift?
- Explain how to handle data deduplication in ETL pipelines.

3. Big Data (Spark, Hadoop, EMR)

- How do you handle skewed data in Spark joins?
- Explain lazy evaluation in Spark.
- What's the difference between persist() vs cache() vs checkpoint() in Spark?
- How would you debug a Spark job running slow on EMR?
- How do you manage small files problem in Hadoop/S3?

4. AWS Data Engineering Services

- Compare AWS Glue vs AWS Data Pipeline.
- How do you design a data lake using Amazon S3, Glue, and Athena?
- What is Kinesis Data Streams vs Kinesis Firehose? When do you use each?
- How do you implement real-time streaming ingestion in AWS?
- What are best practices for securing data pipelines in AWS?

5. Python & Data Structures

- Write a Python function to flatten a nested list.
- Given a log file, parse it and extract the top 10 most frequent IP addresses.
- How do you handle memory-efficient file processing in Python?
- Explain the difference between deep copy vs shallow copy in Python.
- How would you implement a producer-consumer pattern in Python?

6. System Design & Scalability

- Design a system to handle 1B+ daily clickstream events.
- How do you design a data warehouse for Amazon Prime Video analytics?
- What's your approach to designing a real-time recommendation system?
- How would you implement data versioning in a data lake?
- Explain the architecture of an end-to-end ETL pipeline in AWS.

7. Behavioral (Amazon Leadership Principles)

- Tell me about a time you worked on a project with ambiguous requirements.
- Describe a situation where you had to deal with a huge data failure.
- Have you ever challenged a decision made by a manager? What happened?
- Tell me about a time you automated a manual process.
- How do you prioritize when you have multiple conflicting deadlines?

These questions are real and frequently asked in Amazon Data Engineer interviews.

Pro Tip: Amazon interviews are 50% technical, 50% leadership principles. Be ready with STAR (Situation, Task, Action, Result) stories for each principle.